S1 — Lifestyle, Health and Risk

1.1 The Cardiovascular System

Structure of the Heart

The human heart is a muscular double pump consisting of four chambers: two thin-walled atria that receive blood and two thick-walled ventricles that pump blood out. The left ventricle wall is substantially thicker (approximately three times) than the right because it must generate enough pressure to push blood around the entire systemic circulation, while the right ventricle only needs to push blood the short distance to the nearby lungs (pulmonary circulation). The interventricular septum separates the left and right sides, preventing oxygenated and deoxygenated blood from mixing.

> **Cardiac cycle:** The sequence of events in one complete heartbeat, including atrial systole, ventricular systole, and diastole. One complete cycle lasts approximately 0.8 seconds at a resting heart rate of 75 beats per minute.

Key structures include the atrioventricular valves (bicuspid/mitral on the left, tricuspid on the right) which prevent backflow from ventricles to atria, and the semilunar valves in the aorta and pulmonary artery which prevent backflow into the ventricles during diastole. The valves open and close in response to pressure differences between the chambers and vessels. The tendinous cords (chordae tendineae) anchor the AV valves to the papillary muscles of the ventricle wall, preventing the valves from inverting under pressure.

The heart has its own blood supply via the **coronary arteries**, which branch from the aorta just above the aortic semilunar valve. The cardiac muscle is myogenic, meaning it generates its own rhythmic contractions without nervous stimulation. The sinoatrial node (SAN) in the right atrium acts as the natural pacemaker, sending electrical impulses across both atria (causing atrial systole), then to the atrioventricular node (AVN), which delays the signal slightly before passing it down the bundle of His and Purkinje fibres to trigger ventricular contraction from the apex upward.

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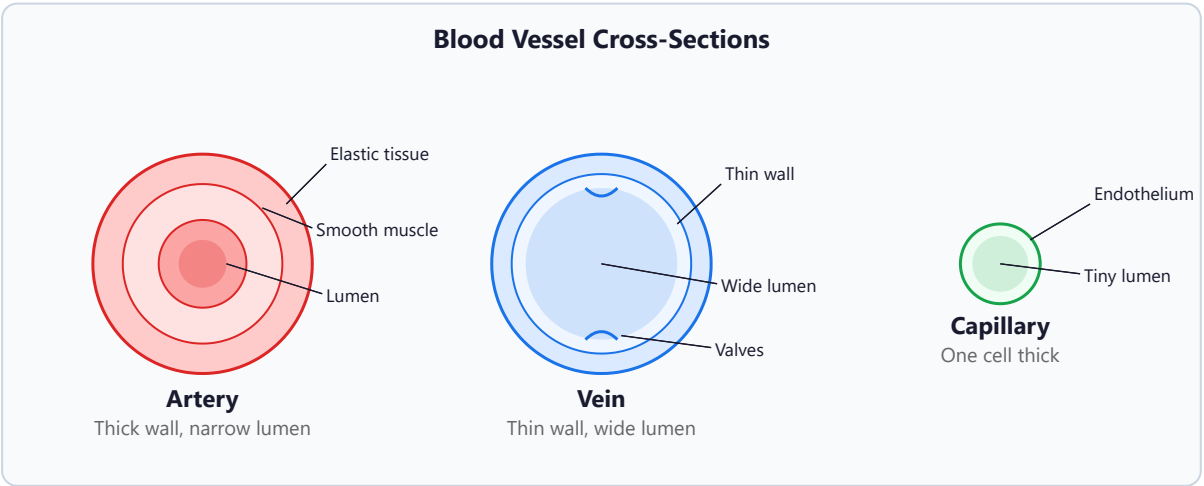
Exam Tip: In questions about the conducting system of the heart, always mention the sequence: SAN → wave of depolarisation across atria → AVN (slight delay to allow atria to empty) → bundle of His → Purkinje fibres → ventricular contraction from the apex. The delay at the AVN is critical for coordinated pumping.

Blood Vessels

Vessel	Wall Thickness	Lumen	Function
Artery	Thick (elastic tissue + smooth muscle)	Narrow, round	Carry blood away from heart at high pressure
Arteriole	Moderate (smooth muscle)	Very narrow	Control blood flow to capillaries; vasoconstriction/dilation regulates blood pressure and distribution
Capillary	One cell thick (endothelium only)	Very narrow (one RBC wide, ~7-8 µm)	Gas and nutrient exchange with tissues; short diffusion distance
Venule	Thin	Small	Collect blood from capillaries; some exchange can occur
Vein	Thin (less elastic tissue and muscle)	Wide, irregular	Return blood to heart at low pressure; contain valves to prevent backflow



Arteries have thick walls with abundant elastic tissue that stretches and recoils to maintain blood pressure between heartbeats (the Windkessel effect). This creates a smooth, continuous flow in arteries despite the pulsatile output of the heart. The elastic recoil converts the intermittent high-pressure surges from ventricular contraction into a more steady flow by the time blood reaches the capillaries. Veins have thinner walls because blood pressure is much lower; they rely on skeletal muscle contraction (the skeletal muscle pump), valves to prevent backflow, the breathing mechanism (pressure changes in the thorax during ventilation), and smooth muscle in the vein walls to return blood to the heart.

Real-world application: Varicose veins develop when the valves in veins become weakened or damaged, allowing blood to pool and the vein to swell. This is most common in leg veins, where blood must be pushed against gravity. Risk factors include prolonged standing, obesity, and pregnancy. Understanding blood vessel structure explains why arteries rarely develop varicosities (their thick walls resist distension) while veins are vulnerable.

> **Exam Tip:** When comparing arteries and veins in exam questions, always mention the relative thickness of the wall, the size of the lumen, the presence or absence of valves, and the amount of elastic tissue and smooth muscle. Link structure to function for full marks — for example, explain *why* arteries have more elastic tissue (to withstand and smooth out high pressure), not just *that* they do.

The Cardiac Cycle in Detail

During **atrial systole** (~0.1 s), the atria contract and blood is pushed through the open atrioventricular valves into the ventricles. This tops up the ventricles, which are already about 70% full from passive filling during diastole. During **ventricular systole** (~0.3 s), the ventricles contract from the apex upward, the atrioventricular valves snap shut (producing the first heart sound, "lub"), and blood is forced through the semilunar valves into the aorta and pulmonary artery. During **diastole** (~0.4 s), the heart muscle relaxes, semilunar valves close (producing the second heart sound, "dub"), and blood flows passively from the veins into the atria.

Pressure changes drive the cardiac cycle. When ventricular pressure exceeds atrial pressure, the AV valves close. When ventricular pressure exceeds arterial pressure, the semilunar valves open. When ventricular pressure drops below arterial pressure (at the start of diastole), the semilunar valves close. Understanding these pressure relationships is essential for interpreting cardiac cycle graphs.

Interpreting cardiac cycle graphs: The x-axis shows time; the y-axis shows pressure. Three lines represent aortic pressure, ventricular pressure, and atrial pressure. Valve events occur at the intersection points: AV valves close when the ventricular pressure line crosses above the atrial pressure line; semilunar valves open

when ventricular pressure exceeds aortic pressure. The volume of blood in the ventricle decreases sharply during ventricular systole (ejection) and increases during diastole (filling).

Exam Tip: When reading a cardiac cycle graph, identify the four key valve events: (1) AV valves close (ventricular pressure > atrial pressure), (2) semilunar valves open (ventricular pressure > aortic pressure), (3) semilunar valves close (ventricular pressure drops below aortic pressure), (4) AV valves open (ventricular pressure drops below atrial pressure). The period between events 1 and 2 is isovolumetric contraction; between 3 and 4 is isovolumetric relaxation.

Cardiac Output

$$\text{Cardiac output (CO)} = \text{Heart rate (HR)} \times \text{Stroke volume (SV)}$$

Cardiac output is the volume of blood pumped by one ventricle per minute, measured in cm^3 per minute (or dm^3 per minute). A typical resting cardiac output is about 5,000 cm^3/min (70 beats/min × ~70 cm^3 per beat). During vigorous exercise, cardiac output can increase to over 20,000 cm^3/min through increases in both heart rate and stroke volume. Heart rate is controlled by the cardiovascular centre in the medulla oblongata, which modulates the SAN via the sympathetic (accelerator) and parasympathetic (vagus) nerves.

1.2 Water and Transport in the Blood

Properties of Water

Hydrogen bond: A weak electrostatic attraction between a slightly positive hydrogen atom (bonded to an electronegative atom such as oxygen or nitrogen — the δ+ end) and a lone pair of electrons on a nearby electronegative atom (δ−). In water, each molecule can form up to four hydrogen bonds with neighbouring molecules, creating an extensive hydrogen-bonded network.

Water is essential for life due to several properties arising from hydrogen bonding:

- **High specific heat capacity** (4.18 J g^{-1} K^{-1}) — a large amount of energy is needed to raise the temperature of water because energy must first break hydrogen bonds before increasing molecular

kinetic energy. This means aquatic environments and body fluids remain relatively stable in temperature, protecting enzymes and metabolic reactions from thermal damage.

- **High latent heat of evaporation** (2,260 J g^{-1}) — evaporation of sweat cools the body effectively because breaking hydrogen bonds requires significant energy, which is absorbed from the skin surface. This is why sweating is such an efficient cooling mechanism.
- **Cohesion and adhesion** — hydrogen bonds between water molecules create surface tension (allowing insects like pond skaters to walk on water) and allow water to be pulled up xylem vessels in plants as a continuous column (the transpiration stream). Adhesion to xylem walls contributes to capillary action.
- **Excellent solvent properties** — the polar nature of water (the oxygen end is δ−, hydrogen ends are δ+) allows it to dissolve ionic compounds (by surrounding ions with hydration shells) and polar substances, making it an excellent transport medium for blood plasma, cytoplasm, and tissue fluid.
- **Maximum density at 4 °C** — ice is less dense than liquid water because hydrogen bonds in ice hold molecules in a rigid, open lattice. Ice therefore floats, insulating water beneath and allowing aquatic organisms to survive in winter.
- **Reactant in metabolic reactions** — water is a substrate in many biochemical reactions, including hydrolysis (breaking bonds with the addition of water) and photolysis (the splitting of water in photosynthesis).

Blood as a Transport Medium

Blood plasma is the liquid component of blood, consisting of about 90% water. It transports dissolved substances including glucose, amino acids, urea, hormones, antibodies, carbon dioxide (as hydrogencarbonate ions, HCO_3^-), and plasma proteins such as fibrinogen (for clotting) and albumin (maintains osmotic pressure and transports fatty acids). Red blood cells (erythrocytes) transport oxygen bound to haemoglobin; white blood cells (leucocytes) are part of the immune response; platelets are cell fragments involved in blood clotting.

Tissue fluid formation: At the arterial end of a capillary bed, the hydrostatic pressure of the blood forces water and small dissolved molecules out through the thin capillary walls into the surrounding tissues (forming tissue fluid). Larger molecules such as plasma proteins remain in the blood. At the venous end, the reduced hydrostatic pressure and the osmotic effect of plasma proteins draw most of the fluid back into the capillary. Excess tissue fluid drains into lymph capillaries and is returned to the blood via the lymphatic system.

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Exam Tip: Be precise about the difference between blood plasma, tissue fluid, and lymph. Plasma is in blood vessels and contains plasma proteins. Tissue fluid is essentially plasma without the large proteins (it has been filtered through capillary walls). Lymph is tissue fluid that has entered lymphatic vessels. The composition differences arise from the filtration process at capillary walls.

1.3 Lipids and Diet

Structure of Lipids

Lipids are biological molecules that are insoluble in water but soluble in organic solvents (such as ethanol and acetone). The main types relevant to health are triglycerides, phospholipids, and cholesterol.

Triglyceride: A lipid formed by the condensation of one glycerol molecule with three fatty acid molecules, producing three ester bonds and releasing three molecules of water. Triglycerides are the main form of energy storage in animals, stored in adipose tissue.

Fatty acids can be **saturated** (no C=C double bonds, maximum hydrogen atoms, straight chains that pack closely together) or **unsaturated** (one or more C=C double bonds, fewer hydrogen atoms, kinked chains that do not pack as closely). Monounsaturated fatty acids have one double bond; polyunsaturated fatty acids have two or more. Saturated fats tend to be solid at room temperature (e.g., butter, lard) because their straight chains pack tightly, forming strong intermolecular forces. Unsaturated fats are liquid at room temperature (e.g., olive oil, sunflower oil) because kinks in their chains prevent close packing.

Trans fats are produced by partial hydrogenation of vegetable oils (an industrial process). They have a similar shape to saturated fats despite containing double bonds, because the hydrogen atoms are on opposite sides of the double bond (trans configuration) rather than the same side (cis configuration). Trans fats are strongly associated with increased CVD risk and have been banned or restricted in many countries.

The Emulsion Test for Lipids

To test for lipids: dissolve the sample in ethanol, then pour into water. A cloudy white emulsion indicates lipids are present (the insoluble lipid forms tiny droplets suspended in water). No cloudiness indicates a negative result.

Cholesterol and Lipoproteins

Cholesterol is a small lipid with a four-ring steroid structure. It is essential for cell membrane structure (it sits between phospholipid tails, regulating fluidity and stability), steroid hormone synthesis (e.g., oestrogen, testosterone, cortisol), bile salt production (needed for fat emulsification in the small intestine), and vitamin D synthesis. However, elevated blood cholesterol levels are a major risk factor for atherosclerosis and cardiovascular disease.

Cholesterol is transported in the blood by lipoproteins because it is insoluble in the aqueous plasma:

Lipoprotein	Composition	Role	Effect on CVD Risk
LDL (Low-density lipoprotein)	High cholesterol, low protein	Transports cholesterol from liver to tissues; delivers cholesterol to cells via LDL receptors	Increases risk — deposits excess cholesterol in artery walls, contributing to plaque formation
HDL (High-density lipoprotein)	Low cholesterol, high protein	Transports cholesterol from tissues back to liver for recycling or excretion in bile	Decreases risk — removes cholesterol from artery walls ("reverse cholesterol transport")

A high LDL:HDL ratio is a significant risk factor for coronary heart disease. Diet (reducing saturated fat, increasing omega-3 fatty acids from oily fish), exercise (increases HDL), genetics (e.g., familial hypercholesterolaemia involves faulty LDL receptors so LDL remains in the blood), and medication (statins inhibit HMG-CoA reductase, the enzyme in the liver that makes cholesterol, thereby reducing blood LDL levels) can all influence this ratio.

Exam Tip: When discussing statins, remember they work by inhibiting cholesterol synthesis in the liver, causing liver cells to up-regulate LDL receptors and remove more LDL from the blood. Common exam distractors: statins do not directly remove cholesterol from plaques; they reduce *new* cholesterol production, which lowers blood LDL over time. Discuss both benefits (reduced CVD risk in high-risk patients) and risks (side effects including muscle pain, liver damage, and the ethical debate about medicating large populations).

Energy and Diet

Energy balance determines whether body mass increases, decreases, or remains stable. If energy intake exceeds energy expenditure, the excess is stored (mainly as fat), leading to weight gain. Energy intake from

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food is measured in kilojoules (kJ) or kilocalories (kcal). Lipids provide more than twice the energy per gram (37 kJ/g) compared to carbohydrates (16 kJ/g) or proteins (17 kJ/g) because lipids have a higher proportion of carbon-hydrogen bonds that can be oxidised and are more reduced (have more hydrogen atoms per carbon atom).

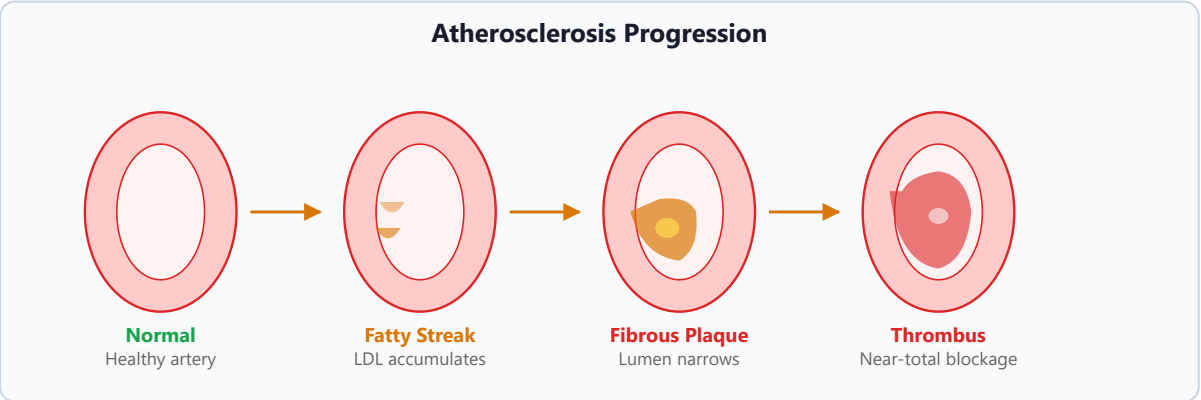
$$\text{BMI} = \text{mass (kg)} \ / \ \text{height}^2 \ (\text{m}^2)$$

A BMI of 18.5–24.9 is considered normal; 25–29.9 is overweight; 30+ is obese. However, BMI has significant limitations: it does not distinguish between fat and muscle mass (a muscular athlete may have a high BMI but low body fat), nor does it indicate fat distribution. Visceral fat (around internal organs, apple-shaped distribution) is more metabolically active and strongly associated with CVD, type 2 diabetes, and metabolic syndrome than subcutaneous fat (under the skin, pear-shaped distribution). Waist-to-hip ratio is often a better predictor of CVD risk than BMI alone.

1.4 Cardiovascular Disease

Atherosclerosis

Atherosclerosis: The progressive build-up of fatty deposits (atheromas or plaques) within the walls of arteries, particularly at points of branching or turbulent flow. This narrows the lumen, restricts blood flow, and reduces oxygen delivery to tissues. It is the underlying cause of most cardiovascular diseases.



The process begins when the **endothelial lining** of an artery is damaged by risk factors such as high blood pressure, toxins from smoking, or high blood glucose. The damaged endothelium triggers an inflammatory

response. White blood cells (monocytes) migrate into the artery wall and become macrophages, which engulf LDL cholesterol that has accumulated beneath the endothelium. These lipid-laden macrophages become **foam cells**, forming a fatty streak. Over time, smooth muscle cells migrate into the area and proliferate, and collagen is deposited, forming a fibrous cap over a lipid core — this is the **fibrous plaque** (atheroma). The plaque narrows the arterial lumen, reducing blood flow and increasing blood pressure. If the fibrous cap ruptures, the exposed lipid core triggers the clotting cascade, forming a **thrombus** (blood clot) that can completely block the artery or break off as an embolus and block a vessel elsewhere.

Coronary Heart Disease and Stroke

If atherosclerosis occurs in the **coronary arteries**, reduced blood flow to the cardiac muscle leads to angina (chest pain during exertion, when oxygen demand exceeds supply) or, if a thrombus completely blocks the artery, a **myocardial infarction** (heart attack). The cardiac muscle downstream of the blockage is deprived of oxygen and glucose, cannot produce ATP by aerobic respiration, and begins to die within minutes. The extent of damage depends on the size and location of the blockage. Treatments include percutaneous coronary intervention (stent insertion) and coronary artery bypass grafts.

A **stroke** (cerebrovascular accident) occurs when blood supply to part of the brain is interrupted, either by a thrombus blocking a cerebral artery (**ischaemic stroke**, ~85% of cases) or by a burst blood vessel causing bleeding into brain tissue (**haemorrhagic stroke**). Brain cells deprived of oxygen die within minutes (neurons cannot respire anaerobically effectively), potentially causing permanent disability (paralysis, speech impairment, memory loss) or death. The specific symptoms depend on which area of the brain is affected.

Blood Clotting (Thrombosis)

When a blood vessel is damaged or a plaque ruptures, the clotting cascade is initiated. This is an enzyme cascade — each step amplifies the signal, allowing a rapid and substantial response from a small initial stimulus:

1. **Platelet plug:** Platelets adhere to exposed collagen in the damaged vessel wall, become activated, change shape, and release clotting factors and ADP (which attracts more platelets).
2. **Clotting cascade:** A series of clotting factors (mostly serine proteases) are activated in sequence. The cascade converts the inactive plasma protein **prothrombin** into the active enzyme **thrombin**.
3. **Fibrin clot:** Thrombin catalyses the conversion of soluble **fibrinogen** into insoluble **fibrin** threads. These threads form a mesh that traps red blood cells and platelets, forming a stable clot that seals the wound.

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Calcium ions (Ca^{2+}) and **vitamin K** are essential cofactors in the clotting cascade. Vitamin K is required for the synthesis of several clotting factors in the liver, which is why the anticoagulant drug warfarin (a vitamin K antagonist) is used to reduce clotting in patients at risk of thrombosis.

Exam Tip: The clotting cascade involves a series of enzyme-catalysed reactions (a cascade amplification). Remember the key sequence: damaged tissue/collagen → platelet activation → prothrombin → thrombin → fibrinogen → fibrin. Questions often ask why clotting involves a cascade rather than a single step — the answer is amplification (a small stimulus produces a large, rapid response) and regulation (multiple steps allow for control and can be inhibited at many points to prevent inappropriate clotting).

1.5 Risk Factors for Cardiovascular Disease

Modifiable Risk Factors

- **Smoking:** Carbon monoxide binds irreversibly to haemoglobin (forming carboxyhaemoglobin), reducing the oxygen-carrying capacity of the blood and increasing heart rate to compensate. Nicotine stimulates adrenaline release, increasing heart rate and blood pressure. Chemicals in smoke (free radicals, tar components) directly damage the endothelium, promoting atherosclerosis. Smoking also increases platelet stickiness and raises LDL levels.
- **High blood pressure (hypertension):** Increases the force on artery walls, causing damage to the endothelium and promoting plaque formation. Defined as systolic >140 mmHg or diastolic >90 mmHg. Often called the "silent killer" because it typically has no symptoms but significantly increases the risk of heart attack, stroke, and kidney damage.
- **High blood cholesterol:** Particularly high LDL and low HDL. Excess LDL deposits cholesterol in artery walls, initiating atherosclerosis. A total cholesterol:HDL ratio >6 is considered high risk.
- **Obesity:** Especially central (visceral) obesity (waist circumference >94 cm in men, >80 cm in women). Increases blood pressure, blood lipid levels, and risk of type 2 diabetes. Adipose tissue is metabolically active and releases inflammatory cytokines that promote atherosclerosis.
- **Physical inactivity:** Regular exercise increases HDL, reduces blood pressure, improves insulin sensitivity, helps maintain healthy weight, strengthens the heart muscle, and improves circulation. Sedentary lifestyles double the risk of CVD.

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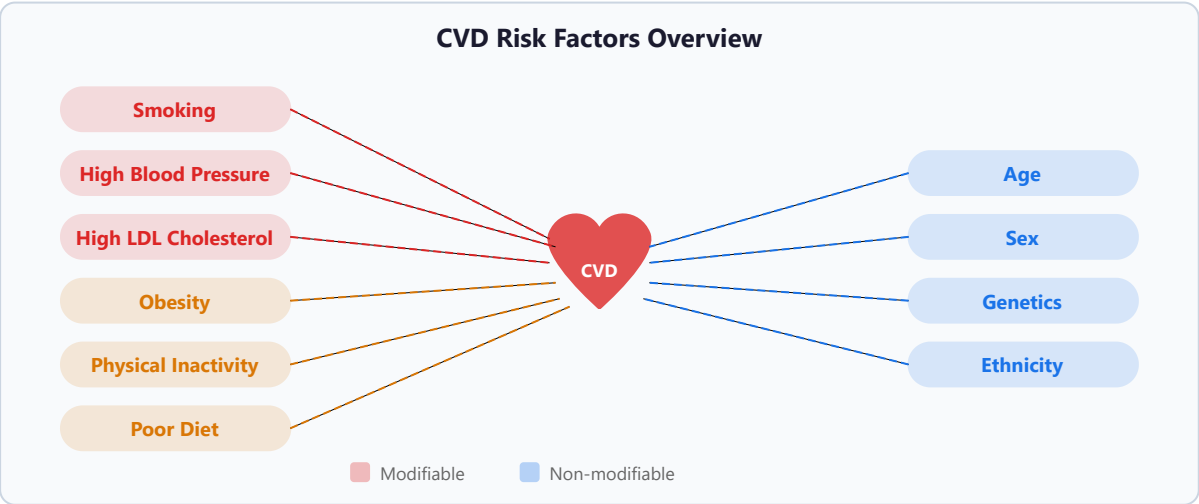
- **Diet:** High saturated fat intake raises blood LDL. High salt intake (sodium) increases blood pressure by increasing water retention (more water in the blood increases blood volume and pressure). Low fibre, low fruit/vegetable intake reduce antioxidant protection. Omega-3 fatty acids from oily fish reduce blood triglycerides and inflammation.
- **Excessive alcohol:** Raises blood pressure and can cause cardiomyopathy (weakening of heart muscle). Heavy drinking increases triglyceride levels. Moderate consumption may have a small protective effect (e.g., red wine polyphenols), but this remains debated.
- **Type 2 diabetes:** High blood glucose damages endothelial cells and promotes glycation of proteins (including LDL), making them more likely to deposit in artery walls. Diabetes approximately doubles the risk of CVD.

Non-Modifiable Risk Factors

- **Age:** Risk increases with age as arteries lose elasticity (arteriosclerosis) and plaques accumulate over decades.
- **Sex:** Males have higher risk before age 50; after menopause, female risk rises as oestrogen levels fall (oestrogen is cardioprotective, helping maintain healthy blood lipid profiles and blood vessel flexibility).
- **Genetics/family history:** Some individuals inherit genes that predispose them to high cholesterol (e.g., familial hypercholesterolaemia, an autosomal dominant condition where LDL receptor genes are mutated so LDL cannot be removed from the blood efficiently), high blood pressure, or type 2 diabetes.
- **Ethnicity:** Some ethnic groups have statistically higher rates of certain cardiovascular conditions — for example, people of South Asian descent have higher rates of coronary heart disease, partly due to genetic factors affecting lipid metabolism and partly due to dietary and lifestyle factors.



Correlation and Causation

Correlation: A statistical relationship between two variables — as one changes, the other tends to change (positive correlation: both increase together; negative correlation: one increases as the other decreases). Correlation does not prove causation; there may be confounding variables, the relationship may be coincidental, or there may be a third variable causing both.

Epidemiological studies (e.g., cohort studies, case-control studies) provide evidence for risk factors. A **causal link** is supported when: the correlation is strong and consistent across multiple independent studies; there is a clear dose-response relationship (the more of the risk factor, the greater the risk); a plausible biological mechanism exists that explains how the factor could cause the disease; the relationship is temporally correct (the risk factor precedes the disease); and the effect is reduced when the risk factor is removed (e.g., CVD risk falls after quitting smoking).

Real-world example: The link between smoking and lung cancer was initially only correlational (from epidemiological studies like those of Doll and Hill in the 1950s). Tobacco companies argued that correlation did not prove causation. However, the causal link was established through multiple lines of evidence: a strong dose-response relationship (more cigarettes = more risk), a plausible biological mechanism (carcinogens in tar cause DNA mutations), temporal correctness (smoking precedes cancer by decades), consistency across populations worldwide, and reduced risk after quitting.

Exam Tip: Always distinguish between correlation and causation in exam answers. Use phrases like "associated with increased risk" rather than "causes" unless a causal mechanism has been established. For full evaluation marks, discuss confounding variables (e.g., smokers may also tend to drink more alcohol or have poorer diets), sample size limitations, whether the study controlled for confounding factors, and the difference between relative and absolute risk.

1.6 Studying Risk and Interpreting Data

Types of Study

Cohort study: A large group of people is followed over time (prospective), and their exposure to risk factors and subsequent health outcomes are recorded. This allows researchers to identify associations between lifestyle factors and disease and to calculate relative risk. The Framingham Heart Study (begun in

1948, still running) is a famous example that identified smoking, high cholesterol, high blood pressure, obesity, and diabetes as major CVD risk factors.

Case-control study: People with a disease (cases) are compared with similar people without the disease (controls), and researchers look back at their past exposure to risk factors (retrospective). These are quicker and cheaper than cohort studies but more prone to recall bias (participants may not accurately remember past behaviours).

Intervention study (clinical trial): Participants are randomly assigned to treatment or control groups to test whether an intervention (drug, lifestyle change) reduces disease risk. Randomised controlled double-blind trials (RCTs) provide the strongest evidence for causation because randomisation controls for confounding variables, and double-blinding eliminates placebo effects and researcher bias. However, they are expensive, time-consuming, and sometimes unethical (you cannot deliberately expose people to harmful risk factors).

Interpreting Data

When evaluating evidence for risk factors, consider: sample size (larger is more reliable and allows detection of smaller effects), whether the study controlled for confounding variables, whether results have been replicated by independent research groups, whether the study has been peer-reviewed, and whether conclusions are actually supported by the data presented. Relative risk and absolute risk are important measures:

$$\text{Relative risk} = (\text{incidence in exposed group}) \,/\, (\text{incidence in unexposed group})$$

A relative risk of 1.0 means no difference in risk; greater than 1.0 means increased risk; less than 1.0 means reduced risk (protective factor). For example, a relative risk of 2.0 for smoking and lung cancer means smokers are twice as likely to develop lung cancer as non-smokers.

Absolute risk is the actual probability of developing a condition over a specified time period (e.g., a 1 in 100 chance of developing heart disease in the next 10 years), while **relative risk** compares probabilities between groups. A large relative risk may correspond to a small absolute risk if the condition is rare. For instance, a factor that doubles the risk of a very rare disease (from 1 in 100,000 to 2 in 100,000) has a relative risk of 2.0 but the absolute risk increase is negligible. This distinction is important for communicating risk to the public and making informed decisions about treatment.

Evaluating Health Claims

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Media reports about health risks often oversimplify or sensationalise scientific findings. Scientists and informed citizens must evaluate whether: the study is published in a peer-reviewed journal (ensuring quality control); the sample size is adequate (small studies may produce unreliable results); confounding variables were controlled (e.g., in a study on wine and heart health, were other lifestyle factors accounted for?); results are statistically significant ($p < 0.05$ means there is less than a 5% probability that the result occurred by chance); the conclusions are supported by the methodology; and whether the study was funded by an organisation with a potential conflict of interest.

Personal anecdotes and testimonials are not scientific evidence. Single studies should be treated with caution until replicated. Systematic reviews and meta-analyses (which combine data from multiple studies) provide the strongest evidence for health claims.

Exam Tip: In data interpretation questions, always refer to specific figures from the data and quote numbers. Calculate percentage changes where appropriate (% change = [new − old] / old × 100). Identify overall trends, anomalies, and suggest explanations for patterns. Remember to discuss reliability (repeatability — can the same researcher get the same results? reproducibility — can different researchers get the same results?) and validity (whether the study actually measures what it claims to measure, with adequate controls for confounding variables).

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